# URBAN SCOUT – UNIFIED MOBILE TOURIST ASSISTANT

CPSC 471 WINTER 2026

GROUP 61

# OUR TEAM

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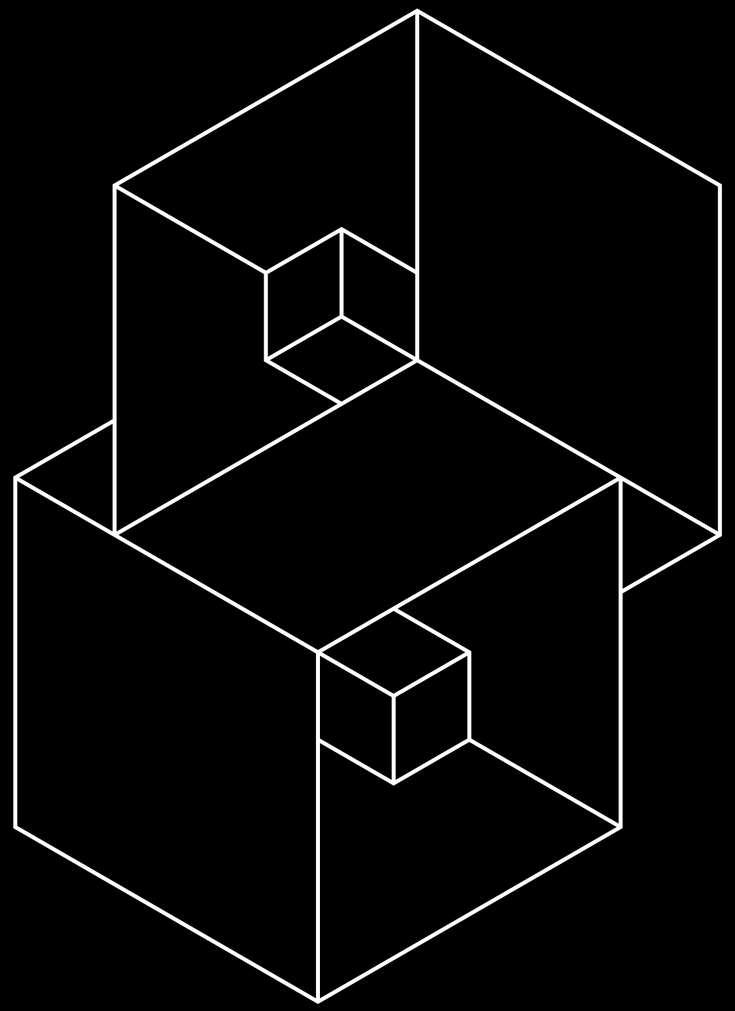
# ABOUT OUR APP

Urban Scout is a mobile-friendly web application that acts as a centralized hub for travel information.
Currently, tourists must use multiple apps such as Google Maps, restaurant review apps, and transit apps to plan their activities. This leads to information fragmentation and app fatigue.
Our system integrates these services into one relational database platform, making travel planning faster and more convenient.



# PROBLEM

Current problems tourists face:

- Information overload
- Data isolation across apps
- Constant context switching
- Lack of awareness about local events

# PROPOSED SOLUTION

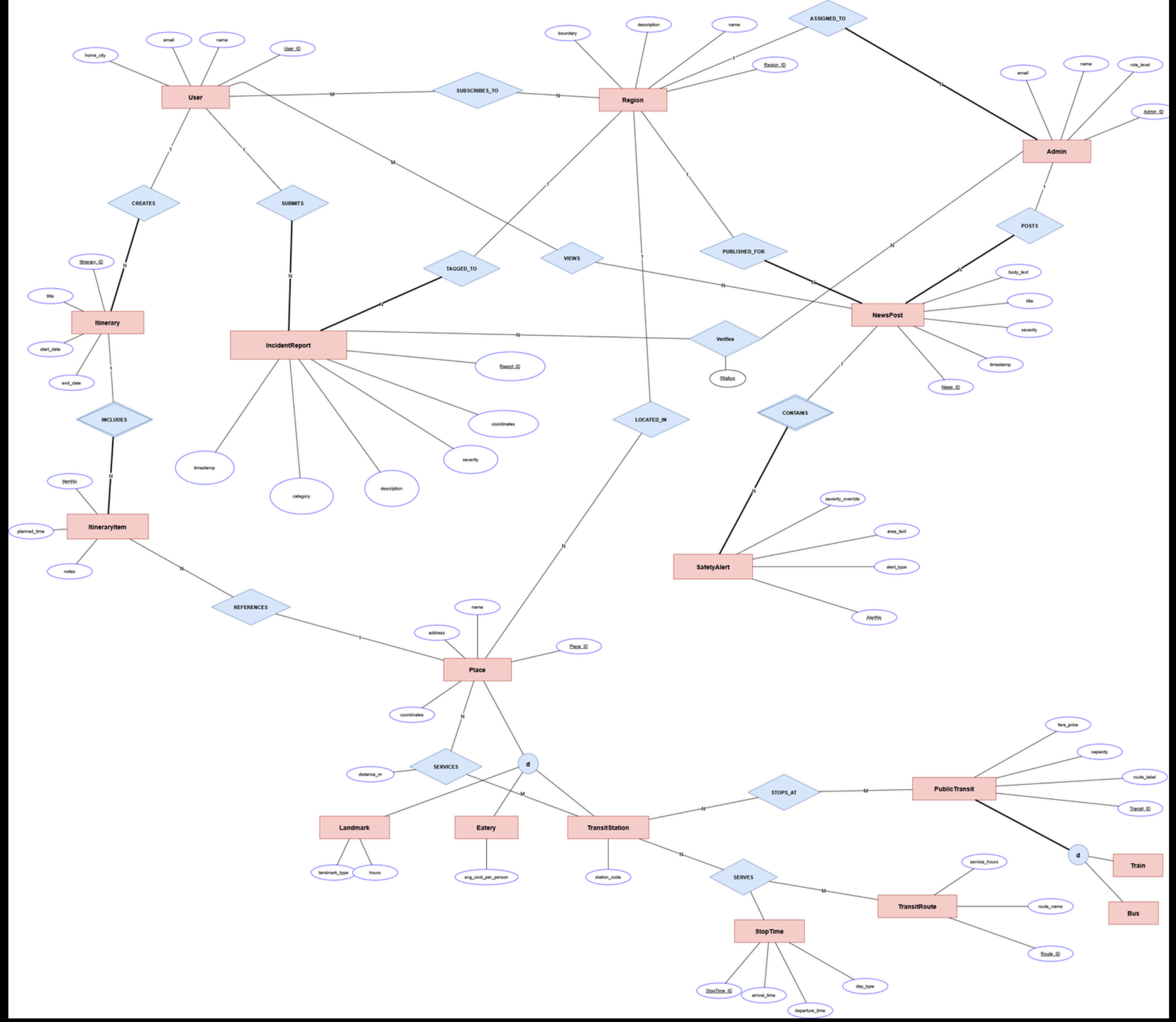
Integrated Travel Database

Landmarks, eateries, and transit stops connected

Local news & safety alerts

Mobile-first interface

# EER DIAGRAM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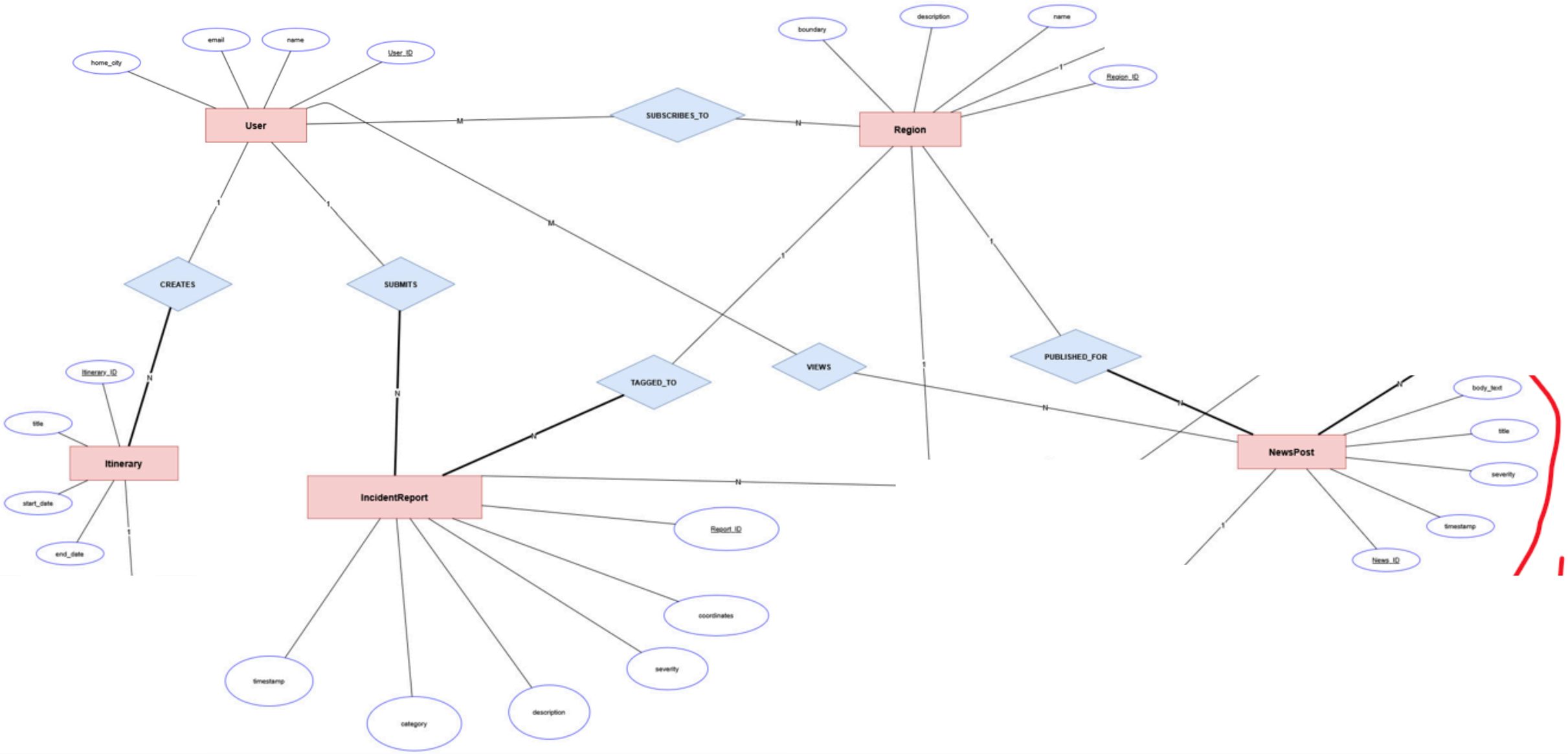
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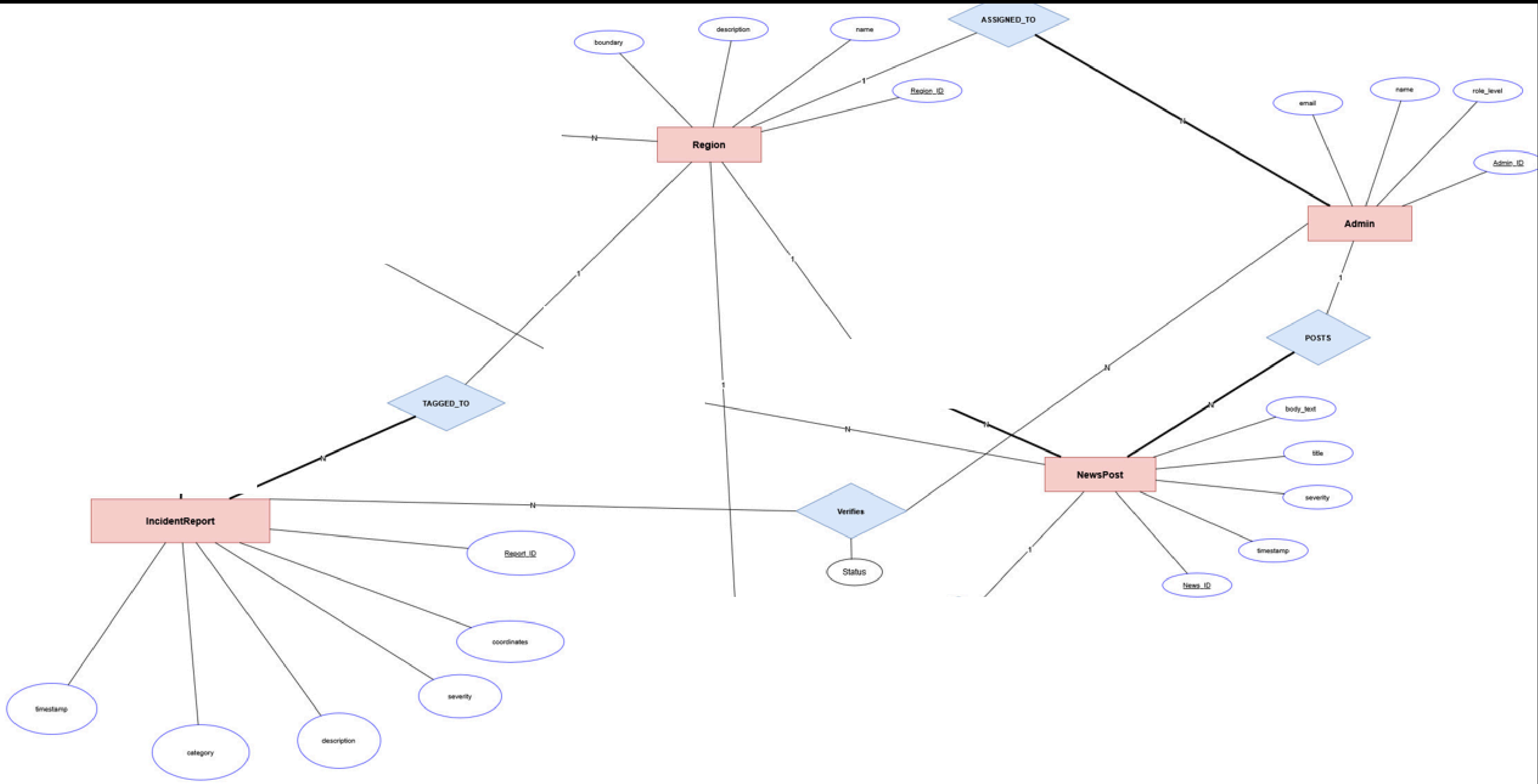
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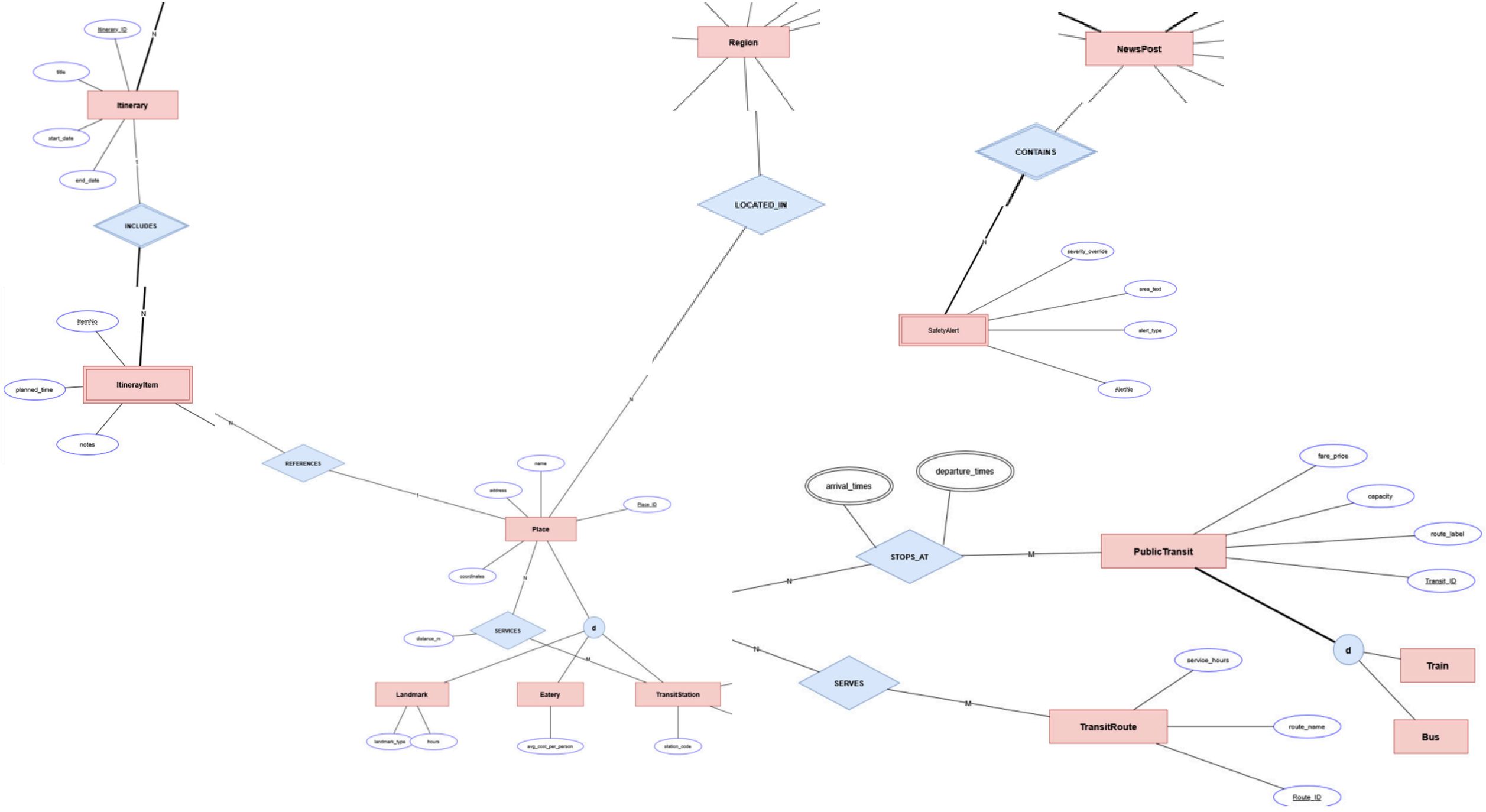
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capacity
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TransitRoute
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route_name
Route_ID
Train
Bus

# Quick Summary of Relations

**User:**

Subscribes to a Region

Creates an Itinerary

Submits an IncidentReport

Views NewsPost

**Admin:**

is Assigned_To a Region

Verifies an IncidentReport

Posts NewsPost

**TransitStation:**

Services **Place**

Serves **TransitRoute**

An **IncidentReport** is Tagged_To a **Region**

A **NewsPost** is Published_For a **Region**

**NewsPost** Contains **SafetyAlert**

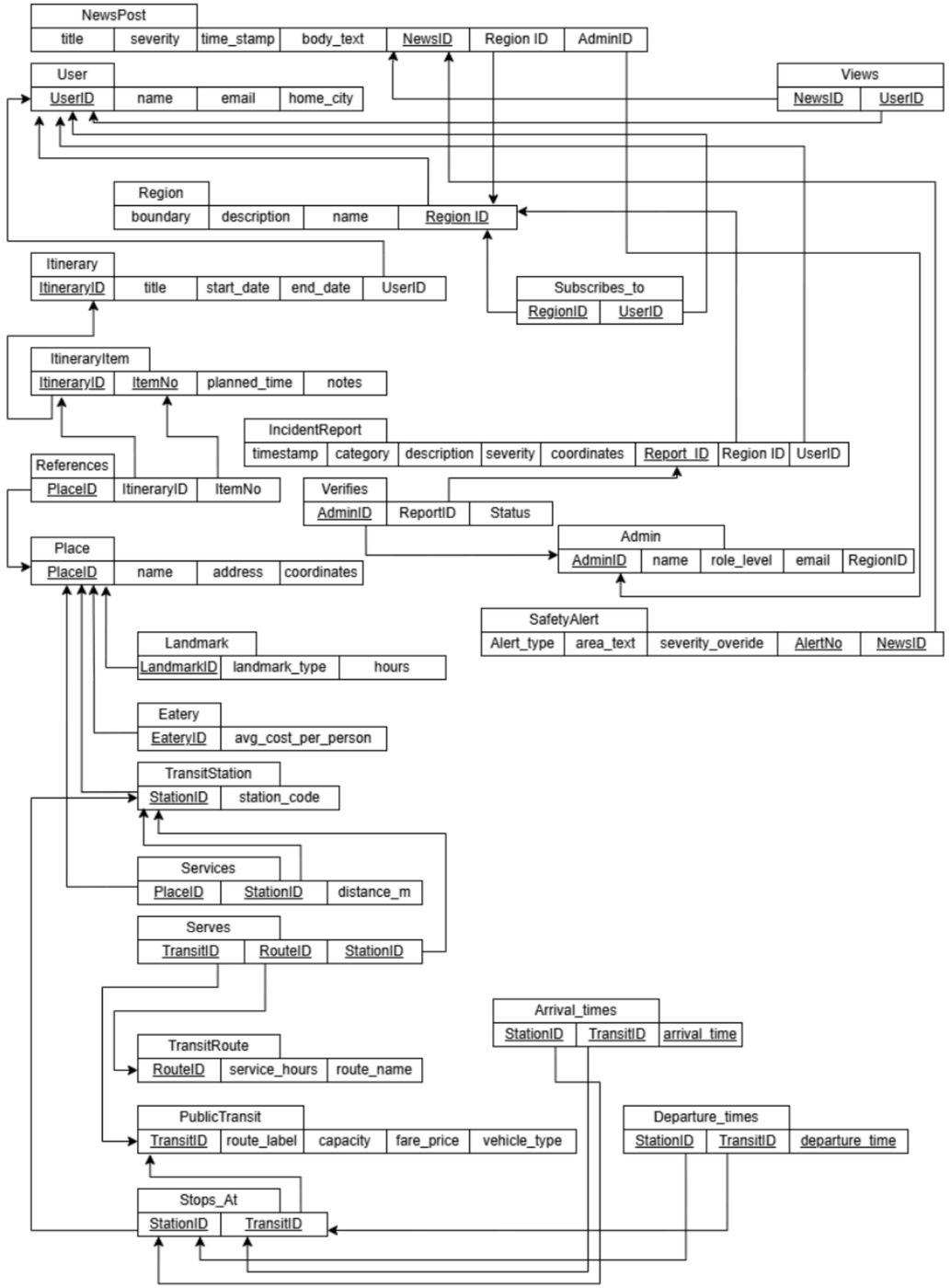
**Itinerary** Includes **ItineraryItem**

**ItineraryItem** References **Place**

**Place** is Located_In a **Region**

**PublicTransit** Stops_At **TransitStation**

# RELATIONAL MODEL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# Keypoints

Separate tables for Landmarks, Eateries, TransitStations

PublicTransit’s vehicle type identified via a bit flag

# PROJECT PLAN & TECH STACK

## Front-End

***React + Vite***

- Component-based UI: Place cards, Itinerary items, News posts, Alert banners
- Vite for fast dev builds (Create React App is deprecated)
- Tailwind CSS for mobile-responsive styling with utility classes and built-in breakpoint prefixes

## Back-End

***Express + Node.js***

- Full-stack JavaScript no language context switching
- Lightweight with full control over routing
- Minimal framework overhead for our project scope

## Database

***MySQL + mysql2***

- Familiar from class and labs TA support available
- Raw SQL queries to demonstrate joins, subqueries, and aggregates
- No ORM — we want to show we understand SQL, not abstract it away
- mysql2 provides prepared statements for SQL injection prevention

# PROJECT PLAN & TECH STACK

## Maps & Coordinates

***Leaflet + react-leaflet***

- Free and open-source (no Google Maps API costs)
- React components for markers and popups
- Schema already stores coordinates for every Place - natural fit

## Authorization

***express-session***

- Two roles: User and Admin with different permissions
- Session-based auth — simpler than JWT for a web-only app
- Single middleware for role-based route protection (admin-only: news, safety alerts)

## Database

***Why These Choices?***

- One language across the stack (JavaScript)
- Tools we know = faster development
- Raw SQL to meet course objectives
- Mobile-responsive for tourists on the go

# FEEDBACK & IMPROVEMENTS

## FEEDBACK RECIEVED

- Clarify entity relationships
- Expand database attributes

## CHANGES MADE

- Improved entity relationships in EER diagram
- Expanded schema with more attributes and entities

# CURRENT PROGRESS & NEXT STEPS

COMPLETED

- PROJECT PROPOSAL
- EER DIAGRAM
- RELATIONAL MODEL

NEXT STEPS

- SYSTEM DESIGN
- IMPLEMENT DATABASE
- DEVELOP BACKEND
- BUILD MOBILE INTERFACE

# THANK YOU